HONGFA "BEN" XUE

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RESEARCH PROFILE

Machine learning researcher and engineer with a PhD in computer engineering and publications in learned code representations, program analysis, symbolic reasoning, and software security. Industry work spans NLP, code translation, LLM based summarization and reranking, model evaluation, distributed PyTorch workflows, and production ML deployment. Interested in reliable language models for software engineering and efficient, verifiable AI systems.

EDUCATION

- **Ph.D., Computer Engineering**, George Washington University, U.S. *Aug 2015 – Dec 2019*
  - **Research Interests:** Machine Learning/Deep Learning, High-Performance Computing, System Security
  - **Thesis:** "Learn2Reason: Joint Statistical and Formal learning Approach to improve the Robustness and Time-to-solution for Software Security"
  - **Advisor:** Guru Prasadh Venkataramani, Professor
- **B.S., Artificial Intelligence,** Beijing University of Posts & Telecom. *Sep 2011 – Jun 2015*

RESEARCH AND PROFESSIONAL EXPERIENCE

- **Machine Learning Engineer II** - Mapbox, CA, U.S. *Nov 2024 – Nov 2025*
  - Evaluated and implemented cloud deployed models for an LLM powered voice assistant and high-performance inference.
  - Built a recommendation engine and a production MLOps pipeline for training, evaluation, versioning, and deployment.
- **Machine Learning Engineer** - Ford Motor Company, CA, U.S. *June 2021 – Sep 2024*
  - Developed neural code translation workflows with data preparation, fine tuning, BLEU and human evaluation, deployment, monitoring, and retraining.
  - Built LLM based diagnostic report summarization and recommendation reranking workflows using GPT 4o, Llama 3.1, and T5.
  - Optimized distributed PyTorch training and inference; managed model versions and checkpoints for reproducible experiments.
- **Data Scientist** - Netskope Inc, CA, U.S. *Jan 2020 – June 2021*
  - Developed BERT based malicious URL classification and optimized inference with ONNX.
  - Built NLP pipelines for sentiment analysis and named entity recognition.
- **R&D Data Science Summer Intern** - Procter & Gamble, Ohio, U.S. *May 2019 – Sep 2019*
  - Developed NLP and topic modeling approaches for short text data using deep learning frameworks.

TECHNICAL SKILLS

Python; PyTorch, TensorFlow, Keras, Hugging Face, scikit learn, spaCy, NLTK, ONNX; NLP, neural machine translation, transformers, LLM fine tuning, distributed training and inference, MLOps, Google Cloud Vertex AI, AWS, Linux, and Unix.

RESEARCH PUBLICATIONS

1. **Hongfa Xue**, Yongsheng Mei, Kailash Gogineni, Guru Venkataramani, and Tian Lan. "Twin-Finder: Integrated Reasoning Engine for Pointer-Related Code Clone Detection." IEEE International Workshop on Software Clones, 2020, London, Ontario, Canada.
2. Yurong Chen, **Hongfa Xue**, Tian Lan, and Guru Venkataramani. "CHOP: Bypassing Runtime Bounds Checking Through Convex Hull Optimization." Computers & Security, 2019.
3. **Hongfa Xue**, Yurong Chen, Guru Venkataramani, and Tian Lan. "AMASS: Automated Software Mass Customization via Feature Identification and Tailoring." EAI Transactions on Security and Safety, 2019.
4. **Hongfa Xue**, Yurong Chen, Guru Venkataramani, and Tian Lan. "Hecate: Automated Customization of Program and Communication Features to Reduce Attack Surfaces." SecureComm 2019, Orlando, FL.
5. **Hongfa Xue**, Shaowen Sun, Guru Venkataramani, and Tian Lan. "Machine Learning-Based Analysis of Program Binaries: A Comprehensive Study." IEEE Access, page 65889-65912, 2019.
6. **Hongfa Xue**, Guru Venkataramani, and Tian Lan. "Clone-Slicer: Detecting Domain Specific Binary Code Clones through Program Slicing." Workshop on Forming an Ecosystem Around Software Transformation, 2018, Toronto, ON, Canada.
7. **Hongfa Xue**, Guru Venkataramani, and Tian Lan. "Clone-Hunter: Accelerated Bound Checks Elimination via Binary Code Clone Detection." ACM SIGPLAN International Workshop on Machine Learning and Programming Languages (MAPL), 2018, Philadelphia, PA.
8. **Hongfa Xue**, Yurong Chen, Guru Venkataramani, Tian Lan, Guang Jin, and Jason Li. "MORPH: Enhancing System Security through Interactive Customization of Application and Communication Protocol Features." In Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security (CCS '18).
9. **Hongfa Xue**, Yurong Chen, Fan Yao, Yongbo Li, Tian Lan, and Guru Venkataramani. "SIMBER: Eliminating Redundant Memory Bound Checks via Statistical Inference." Proceedings of 32nd International Conference on ICT Systems Security and Privacy Protection (IFIP SEC), 2017, Rome, Italy.
10. Fan Yao, Yongbo Li, Yurong Chen, **Hongfa Xue**, Tian Lan and Guru Venkataramani, "StatSym: Vulnerable Path Discovery through Statistics-Guided Symbolic Execution." 2017 47th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), Denver, CO, USA, 2017.
11. Kailun Zhang, Mingyin Wang, Xiaoyue Cong, Fang Huang, **Hongfa Xue**, Lei Li, Zhiqiao Gao, "Personal Attributes Extraction Based on the Combination of Trigger Words, Dictionary and Rules." In Proceedings of the Third CIPS-SIGHAN Joint Conference on Chinese Language Processing, pages 114–119, Wuhan, China. Association for Computational Linguistics.